

Enhanced electrochemical properties of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ via lithium boron oxide glass surface treatment

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ABSTRACT

A $\text{Li}_2\text{O} \cdot 2\text{B}_2\text{O}_3$ layer is successfully coated on the surface of Li-rich layered cathode $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$, which is demonstrated by X-ray diffraction (XRD), scanning electron microscopy (SEM) and transmission electron microscopy (TEM). The electrochemical performance of the pristine and LBO-coated materials have been investigated by galvanostatic charge/discharge. The experiment results indicate that the electrochemical performances of Li-rich cathode materials are obviously improved by LBO surface modification. Particularly, the sample with 3 wt.% LBO-coating shows the best cycling performance, with capacity retention of 92% after 100 cycles, which is much better than the pristine one. Electrochemical impedance spectroscopy (EIS) reveals that the enhanced electrochemical performances of the LBO-coated sample are due to the enhancement of surface electronic conductivity and the suppression of the decomposition of electrolyte.

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1. Introduction

With the development of hybrid electric vehicles (HEVs) and electric vehicles (EVs), the high-energy and high-power Li-ion batteries (LIBs) are in great demand [1–5]. As compared with the conventional cathode materials such as layered LiCoO_2 [6] and olivine LiFePO_4 [7], Li-rich layered-oxide positive electrode materials are more attractive due to the higher theoretical specific capacity as large as 250 mAh g^{-1} and relative low cost. The successful application of these materials can increase the energy density of Li-ion batteries, such as $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$.

Nevertheless, the Li-rich Mn-based oxide materials have several shortcomings [8], including poor rate capacity and insufficient capacity retention, which greatly impeded their actual applications. Various effective methods have been adopted to solve these critical problems via coatings of sundry materials [9–13], such as metal oxides [14,15], metal phosphates [16,17], and metal fluorides [18]. A proper coating layer can defend the electrode surface from direct contact with the electrolyte and reduce the side-reactions between the electrolyte and electrode [19,20].

Lithium boron oxide ($\text{Li}_2\text{O} \cdot 2\text{B}_2\text{O}_3$, or LBO) glass is typically solid lithium ionic conductor with good lithium ionic conductivity and electrochemical stability against high oxidation potential. In addition, good wetting properties and relatively low viscosity of the LBO allow easy processing and result in even coverage with the use of a minimal amount of materials [21,22]. Based on the above statements, the

presence of LBO at the surface of host materials might help to improve cycling stability and the rate capability even at high voltage. Also this work has not yet been reported, so LBO glass has been employed as a suitable coating for $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$.

2. Experimental

2.1. Powder preparation

The bare $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ powder was synthesized by oxalate co-precipitation method [23] and annealing at 950°C in air for 16 h. To prepare sample coated with LBO, stoichiometric $\text{LiOH} \cdot \text{H}_2\text{O}$ and H_3BO_3 (1:2 in molar) was dissolved in alcohol under gently stirring. Then, Bare $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ powder was added into the solution stirred continually, and heated at 80°C until the solvent completely evaporated. In the end, the remaining powder was calcined in air at 500°C for 10 h to obtain the sample. For this research, the $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ powder was coated with different amounts of LBO glass in the range of 1 to 5 wt.%. We defined the LBO-glass-coated composites with coating treatment of 0 wt.%, 1 wt.%, 2 wt.%, 3 wt.%, and 5 wt.% as LBO0, LBO1, LBO2, LBO3, and LBO5 respectively.

2.2. Physical characterizations

The powder X-ray diffraction (XRD, Rint-2000, Rigaku, Japan) using $\text{Cu K}\alpha$ radiation was employed to identify the crystalline phase of the synthesized materials. The morphology of the composites was measured by scanning electron microscopy (SEM, Sirion 200), and by transmission electron microscope (Tecnai G12, 200KV).

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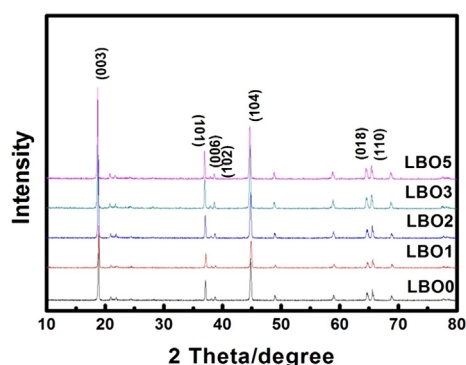


Fig. 1. XRD patterns of LBO0, LBO1, LBO2, LBO3, LBO5.

2.3. Electrochemical tests

The electrochemical characterizations were detected by using a standard CR2025 coin-type cell composed of the cathode, metallic lithium anode, a commercial polypropylene micro-porous separator, and 1 mol/L LiPF_6 in EC, EMC and DMC (1:1:1 v/v/v). The samples were prepared by coating homogenous slurries on an aluminum foil. The slurry was mixed with 10% conductive-carbon, and 10% PVDF binder was coated onto an aluminum foil current collector and dried for 6 h at 120 °C. Then it was cut into the shape of circular disks and kept in a vacuum drying oven prior to use. The assembly of the cells was executed in an argon-filled glove box. The cells are charged and discharged over a voltage range of 2.5–4.6 V versus Li/Li^+ electrode at current density of 0.2 C, 0.5 C, and 1 C (250 mA g^{-1}) at room temperature.

3. Results and discussion

XRD patterns of pristine and LBO coated samples are shown in Fig. 1. Except for the weak peaks located at $2\theta = 20\text{--}25^\circ$, the weak peaks

located are caused by the short-ranged superlattice ordering of the Li, Ni and Mn atoms in the transition metal layers [24,25]. The other peaks in the XRD pattern of LBO0 can be indicated to the $\alpha\text{-NaFeO}_2$ structure with space group of R-3m. In addition, strong and narrow diffraction peaks as well as clear peak splits of (006)/(102) and (018)/(110) reveal the highly ordered layered structure of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ powders. The diffraction peaks of the modified samples (LBO1–LBO5) show almost no differences with those of the pristine material, indicating that the LBO coating layer has little influence on the bulk structure of the bare sample. Since no other diffraction peaks show up, it is believed that amorphous LBO glass is well coated onto the surface of the pristine material.

In order to confirm further information of the coating layer on the materials, SEM and TEM were used to study surface morphologies and microstructures of these samples. Fig. 2(a) and (b) shows SEM images of LBO0 and LBO3 samples which exhibit similar surface morphologies. The particle size of LBO3 remains unchanged and there is less agglomeration after modification. Fig. 2(c) and (d) shows TEM images of LBO0 and LBO3 particles. An apparent thin coating layer with a thickness of approximately 8 nm is observed on the surface of LBO3. This phenomenon suggests that the coating layer has been formed on the surface of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ powders. The EDS images of LBO3 are shown in Fig. 3 and confirm that the B elements are distributed in the samples homogenously.

The initial discharge curves of all samples in the range of 2.5–4.6 V at a current density of 12.5 mA g^{-1} is shown in Fig. 4(a). The initial discharge capacities of the samples are 255, 240, 232, 228, 213 mAh g^{-1} , respectively. The decrease of capacity means the LBO glass in the LBO coated samples is a non-electroactive species. Fig. 4(b) is the influence of LBO coating layer on the capacity retention of Li-rich cells in the range of 2.5–4.6 V at 0.2 C, 0.5 C and 1 C (250 mA g^{-1}) rates, respectively. Apparently, pristine Li-rich electrode suffers fast capacity decay during the cycle process. At the current density of 250 mA g^{-1} , it displays an initial discharge capacity of 150 mAh g^{-1} , which sharply decreases to 106 mAh g^{-1} after 100 cycles, with a capacity retention of only 70%. As can be clearly observed in this figure, LBO surface modification greatly improves the cycleability of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$. In

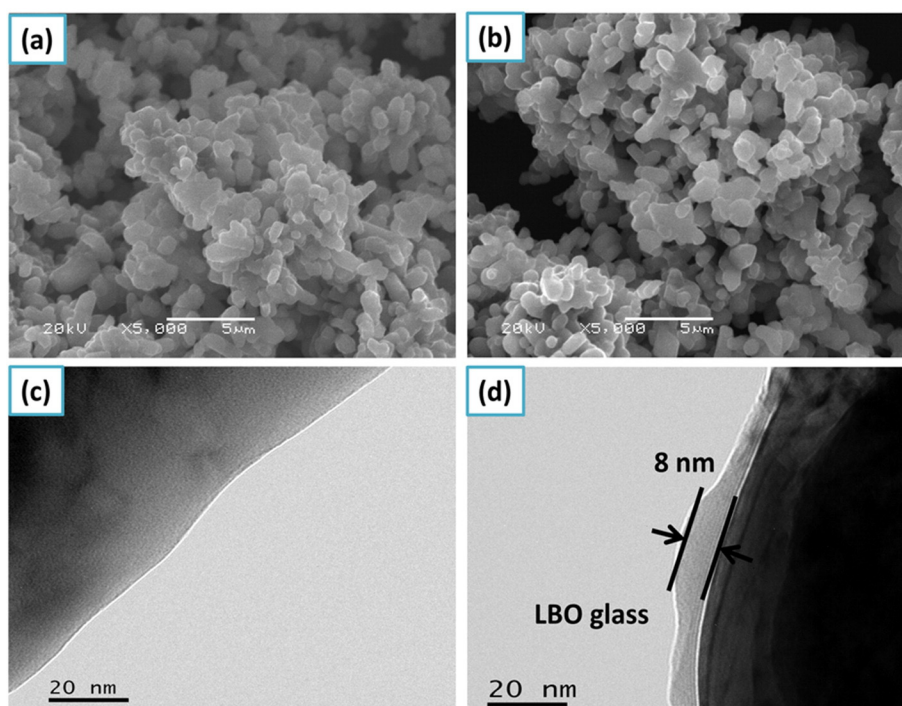


Fig. 2. SEM images of LBO0 (a) and LBO3 (b), TEM images of LBO0 (c) and LBO3 (d).

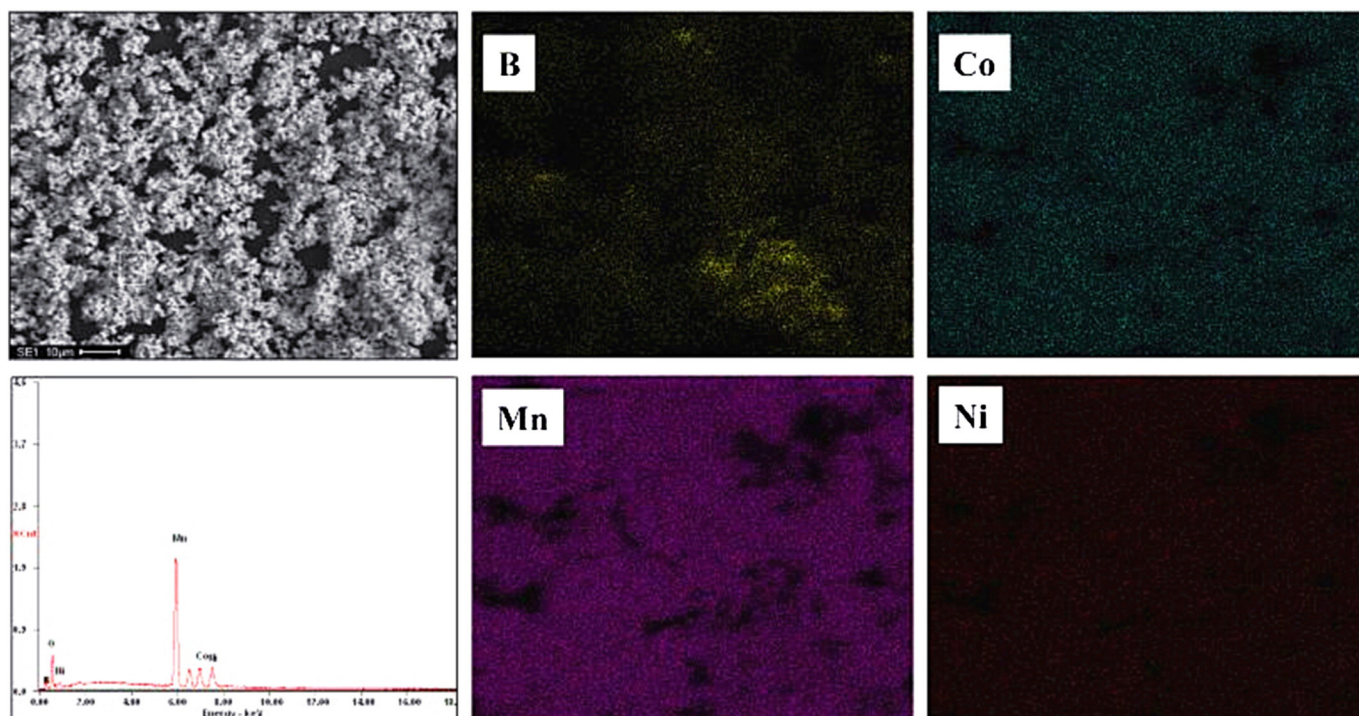


Fig. 3. EDS images of LBO3.

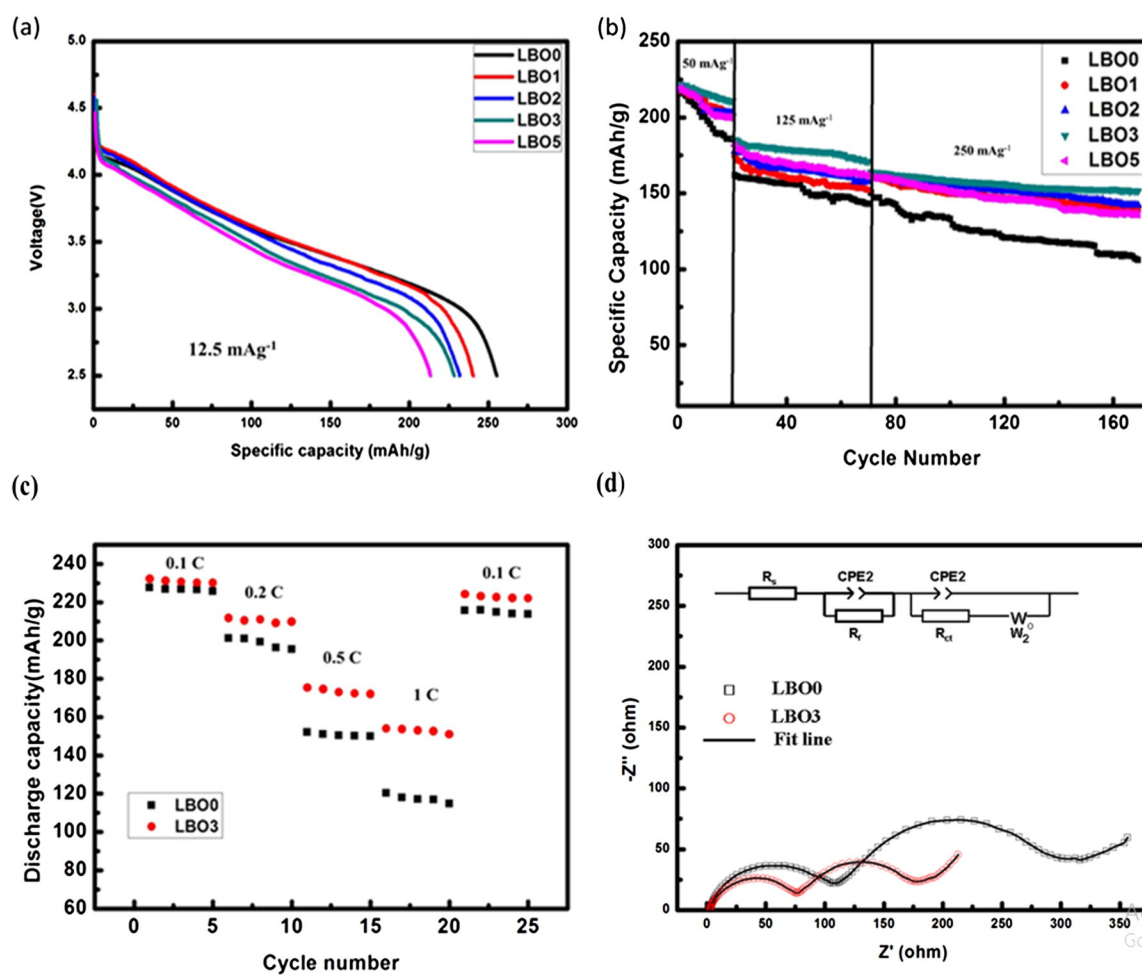


Fig. 4. (a) Initial discharge curves of LBO0, LBO1, LBO2, LBO3, LBO5 at the rate of 0.05 C. (b) Cycleabilities of LBO0, LBO1, LBO2, LBO3, LBO5. (c) Rate performances of LBO0 and LBO3 in the voltage range 2.0–4.6 V at room temperature. (d) Nyquist plots of LBO0 and LBO3 after 3 cycles.

Table 1

The impedance parameters of LBO0 and LBO3.

Sample	R_s (Ω)	R_f (Ω)	R_{ct} (Ω)
LBO0	3	94	222
LBO3	2	69	110

comparison, LBO3 sample exhibits the most excellent cycling performance. The initial discharge capacity is about 164 mAh g^{-1} at the rate of 1 C, which still remains 151 mAh g^{-1} after 100 cycles, with an improved higher capacity retention of 92%. At the current density of 50 mA g^{-1} and 125 mA g^{-1} , the cycling performance of samples are also improved by the LBO surface modification. The improvement can be attributed to the LBO modification layer, which not only prevents the direct contact between the electrode and electrolyte but also stabilizes the cathode structure by decreasing the loss of oxygen [26]. Fig. 4(c) shows the rate performances of pristine and LBO3 sample at 0.1 C, 0.2 C, 0.5 C and 1 C rates. It reveals that the LBO3 sample shows much higher discharge capacities than those of pristine electrode. Moreover, when the current density returns to 0.1 C, the discharge capacity can be almost recovered. All of the electrochemical results indicate that the LBO3 sample has ideal electrochemical reversibility and structural stability.

To further demonstrate the electrode process kinetic behavior, EIS of the LBO0 and LBO3 were conducted on coin cells at open circuit voltage after 3 cycles at 0.1 C. As shown in Fig. 4(d), the obtained plots are composed of two depressed semicircles. The semicircle in the high-frequency region is associated with the surface film resistance (R_f) formed by the decomposition of the electrolyte in the high-voltage range. The semicircle in the middle frequency region represents the charge-transfer resistance (R_{ct}) due to the Li insertion reaction [27,28]. Based on the equivalent circuit given in the inset of Fig. 4(d), the value of R_f and R_{ct} are simulated. It can be seen that LBO coating decreases both R_f and R_{ct} , indicating an enhancement in the kinetics of lithium-ion diffusion through the surface layer and charge transfer reaction and a consequent increase in rate capability. From Table 1, the value of R_f decreases from 94 to 69Ω and R_{ct} decreases from 222 to 110Ω when the coating amount of LBO increases from none to 3 wt.%. The LBO coating decreases the charge-transfer resistance and promotes the charge-transfer reaction in the active electrode, which is helpful for battery property.

4. Conclusions

LBO is coated on the layered $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ electrodes. Electrochemical data reveals that the cycling performance and rate capability are improved by LBO coating. The 3 wt.% LBO-coated material exhibits the optimal performance. The charge–discharge capacity and EIS measurements reveal that the improved electrochemical performance of the LBO-coated sample is due to the enhancement in surface electronic conductivity and the suppression of the SEI layer. The results of above demonstrate that LBO surface coating is an effective application in electrode material design and optimization for LIBs.

Acknowledgments

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